Influenza A in the Air

Influenza A virus · airborne transmission · detection

A knowledge base on airborne transmission, detection and control, compiled as a contribution to the iGEM community

SZPU-China · 2026 iGEM

A contribution to the iGEM community: questions on influenza A in the air, summarised from the literature and released for learning.

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Preface

After our social practice, we recognised that many important questions about influenza A virus in the air — how it is transmitted, how long it can persist in air and on surfaces, how environmental conditions affect it, and what different detection signals can actually establish — draw on several fields and are dispersed across the literature. We therefore read the relevant studies and organised what we learned into this Influenza A in the Air contribution, bringing together the evidence, key data, study conditions and references needed to examine these questions. The document does not present new experimental findings; it structures existing research into a continuous, accessible and traceable resource that can help readers develop an overall understanding of airborne transmission, environmental persistence, detection and control of influenza A. In this way, it provides a reference for learning, discussion and project development, reduces duplicated searching, and makes the material available for further use, checking and improvement within the iGEM community.

Abstract

This document collects what we learned about influenza A virus (IAV) while designing an airborne biosensor, and sets it out in a form that other iGEM teams can use. It began as background reading for our own project. Our social practice, which brought us into contact with clinicians, public-health staff and other iGEM teams, kept producing the same questions: how long infectious virus survives once it leaves a person, what temperature and humidity do to it, where the boundary between droplets and airborne particles should be drawn, and what a positive signal actually proves. None of these questions can be answered from a single interview, and none of them are answered in one place in the literature.

This document is organised around twenty-one questions, each answered from the literature: what is known, the numbers that matter for design decisions, and the references that support them. It follows the questions in the order a team would ask them, explains why each answer matters, and notes where the evidence is thin. It covers the virion and its genome, environmental persistence in air and on surfaces, transmission routes, receptor biology and host range, detection methods and what they measure, the antibody landscape, antivirals and vaccines, and the clinical comparisons that position influenza against COVID-19 and respiratory syncytial virus. Every substantive claim is attributed; where studies disagree, we say so rather than choosing a side.

The document is written for judges and for teams. For judges, it is the record of a contribution: what we read, how we judged it, and what we produced. For teams, it is intended to be reusable — its sections can be lifted into a wiki page, a safety form or a presentation, and the final chapter explains how.

1. Introduction: the question behind our project

Our team is building a yeast-based biosensor for influenza A virus in air. The device has to answer three questions in sequence: can the virus be captured from the air in the size range where it actually travels; can a living cell recognise it specifically; and can the resulting signal be read in a way that tells an operator something true about the sample. Each of these questions is biological, but none of them is answerable from biology alone. The first depends on aerosol physics and on how long the virus stays infectious. The second depends on the structure of the virion and on how antibodies and receptors engage it. The third depends on what our assays measure — infectivity, nucleic acid, or something in between.

When we started, we assumed the literature would settle these questions quickly. It did not. The evidence is spread across virology, aerosol science, clinical infectious disease and engineering, and the papers that address one part often take the others for granted. A study of aerosol stability reports relative humidity but not droplet size; a clinical review reports transmission routes without stating what fraction of exhaled particles falls below the airborne threshold; a modelling paper assumes an infectious dose that no experiment has established. Reading across these literatures requires holding several kinds of uncertainty at once.

The social practice changed how we handled that. Interviews with clinicians and public-health staff did not give us the answers — they gave us better questions, and a clearer sense of which questions a future user of our device would actually ask. A hospital clinician wants to know whether a positive reading means a patient is infectious. A public-health officer wants to know whether a room is safe to reopen. Both questions turn on the same distinction between what a test detects and what is happening in the patient or the room, and neither is answered by a detection limit alone.

After the social practice we decided to stop treating this reading as private preparation. We reorganised it around twenty-one questions, checked each answer against its primary sources, and wrote the result as the document that follows. The contribution is not a new finding. It is a structured account of what is known, where it comes from, and what it implies for anyone trying to detect influenza A in the air.

1.1 Why the questions had to be collected

A team that wants to build an air monitor faces a scattered literature and a short timeline. The questions are predictable — every team working on airborne pathogens asks some version of them — but the answers are not collected anywhere. Reviews exist for each subfield, yet they are written for specialists and assume the reader already knows why droplet size matters or what a plaque assay measures. What is missing is an account written from the position of someone who has to make a design decision this month.

We also found that the same questions recur in conversations with stakeholders, in a different vocabulary. A clinician asks about infectiousness; a physicist asks about aerodynamic diameter; a judge asks what the team understands about its own target. Underneath the different vocabularies is one problem: connecting a measurement to a biological state. This document is organised around that connection.

1.2 How this document is organised

Chapters 2 to 4 establish the problem: the burden of influenza, the physics and epidemiology of transmission, and the structure of the virion. Chapters 5 to 8 follow the virus through its environment and into a detection system: persistence, entry and host range, measurement, and the antibodies used to recognise it. Chapters 9 to 11 place detection in a clinical and public-health context: treatment, the clinical picture, and control measures. Chapters 12 to 14 describe how this document was built, how it can be reused, and what we still do not know.

Each chapter ends with a short section on what the evidence means for a detection device. These sections are our interpretation rather than the authors', and they are marked as such. Every claim can be traced through the reference list to the study that supports it,

2. Influenza A as a public-health problem

Influenza is not a rare disease that occasionally surprises a health system; it is a recurring one that health systems plan around, and its cost is measured both in seasons and in pandemics. The same virus that produces a familiar winter illness in most people produces severe disease in a minority, and the size of that minority changes with the strain, the population and the year. This chapter sets out what the clinical and epidemiological literature says about that pattern, because a detection device is only useful if it changes what someone can do about it.

2.1 Seasonality and the limits of vaccination

Vaccination is the mainstay of influenza control, and the literature is consistent that it reduces morbidity and mortality at the population level. It is equally consistent that its performance varies. Reported effectiveness against medically attended influenza in the United States ranged from about 10% to 60% between 2004 and 2021, with the variation driven by the match between circulating and vaccine strains, by the population studied, and by the season [1]. A tool whose performance swings by a factor of six across years creates a specific kind of uncertainty for public health: the same recommendation carries very different weight from one winter to the next.

Seasonality itself remains incompletely explained. Temperature and humidity change together over the year, human behaviour moves indoors, and school terms reorganise contact patterns, so separating the contributions is difficult. The aerosol literature offers a mechanistic route through humidity, which we take up in Chapter 3 and Chapter 5. For the purposes of this chapter, the point is that influenza arrives predictably in time but unpredictably in severity, and that surveillance has to be both continuous and local.

2.2 The 2009 H1N1 pandemic as a case study

The 2009 H1N1 pandemic is the most completely documented recent example of a novel influenza A strain entering a population with little pre-existing immunity. Two findings from that literature matter for how we think about detection. First, person-to-person transmission appeared similar to that of seasonal influenza, but the relative contributions of small-particle aerosols, large droplets and fomites could not be resolved [2]. Second, roughly one quarter to one half of patients who were hospitalised or died had no reported underlying medical condition [2].

The second finding is the one that changes practice. Risk-group lists — the elderly, pregnant women, people with chronic disease — are useful for targeting vaccination, but they do not identify who will become severely ill. If a substantial share of severe cases occurs in people who would not have been flagged in advance, then population-level detection and rapid access to treatment carry more of the burden than risk stratification alone. That is an argument for measuring the environment, not only the patient.

2.3 Clinical course and why timing matters

Influenza typically produces fever, myalgia, headache and respiratory symptoms, with the most severe symptoms in the first week of illness. Complications include viral pneumonia, secondary bacterial

infection, and — less widely recognised — central nervous system involvement. Febrile seizures are common in infants and children with influenza and are an important cause of hospitalisation; influenza-associated encephalitis and encephalopathy are rarer but more serious [3,4,5]. A device that detects virus in a room does not diagnose any of this, but it can shorten the interval between exposure and the decision to test a patient.

That interval matters because antiviral treatment is time-dependent. In a 24-hospital network in the United States during the 2022–2023 season, adults hospitalised with influenza who started oseltamivir on the day of admission were less likely to progress after admission — 11% versus 21% for an escalation of one level on a pulmonary severity scale — and had lower odds of ICU admission, with an adjusted odds ratio of 0.25 [6]. The effect is real but modest in absolute terms; it is largest when treatment begins early, which is precisely when a detection system could help.

2.4 What this means for a detection device

Three conclusions follow. First, the value of detection is in time saved, not in diagnosis: a device that shortens the path to a confirmatory test is useful even if it never replaces one. Second, the population that benefits is wider than the conventional risk-group list, which means the device has to work in ordinary spaces — classrooms, wards, transport — and not only in clinical settings. Third, because influenza severity varies by season and strain, a detection system needs to be interpretable against a changing baseline; a signal that was unusual in one winter may be ordinary in another.

3. Transmission: droplets, aerosols and fomites

Transmission is where the physics of the virus meets the epidemiology of the disease. For a detector, this is the chapter that decides what to sample and where. For a reader assessing this document, it is also the chapter where the literature is least settled, because the categories used in clinical guidance are not the categories that particle physics produces.

3.1 The classical dichotomy and the 5 µm line

Aerosolised transmission is conventionally divided into droplet and airborne routes. Droplet transmission refers to expelled particles that settle quickly, usually within about a metre of the source, so that infection depends on close proximity. Airborne transmission refers to particles that remain suspended long enough to be inhaled at distance. The World Health Organization applies a 5 µm cut-off, treating particles of 5 µm or less as airborne and larger particles as droplets [7]. The same review is careful to note that size is a continuum: particles on either side of the cut-off can behave in overlapping ways, and the categories describe tendencies rather than distinct mechanisms.

For a design team this matters immediately. If the target is defined by a cut-off, then the inlet, the filter and the elution step all inherit that definition. A sampler that captures only particles above 5 µm will miss the fraction that travels farthest; one that captures everything will collect dust and debris that complicate downstream detection. The choice is not technical housekeeping — it is a decision about which transmission route the device is meant to monitor.

3.2 Droplet physics: evaporation and settling

The size of a droplet at the moment it leaves the respiratory tract is not the size it has when it reaches a sampler. Respiratory droplets are released at close to saturation and evaporate almost immediately in unsaturated air. A 20 µm droplet halves in diameter in less than half a second at 20 °C, so equilibrium size is reached almost instantaneously [8]. What remains — the droplet nucleus — is what travels, and its size depends on ambient relative humidity.

The consequences are quantitative and large. Below about 64% relative humidity, a 10 µm droplet evaporates to roughly 1.9 µm, and its settling time lengthens from about eight minutes to more than three hours [8]. Modelling of a residential room predicts that infectious virus concentrations ten minutes after release are about 2.4 times higher at 10% relative humidity than at 90% [8]. The same modelling makes clear why absolute humidity is a poor sole predictor: it is physically collinear with temperature, so a study that varies one necessarily varies the other.

Droplet size distributions at the source are broad. Pooling seven studies of cough-generated droplets, the geometric mean diameter is about 12.9 µm with a geometric standard deviation of 2.3 [8]. A single cough therefore produces both droplets that settle within a metre and nuclei small enough to remain airborne for hours; the same exhalation contributes to both transmission routes at once. This is the strongest argument against treating the 5 µm line as an epidemiological boundary rather than a modelling convention.

3.3 Aerosolized fomites: dust as a vehicle

A third route has received less attention: virus attached to particles that are not respiratory droplets at all. In a guinea-pig model, infected animals contaminated fur, ears, paws and cage surfaces with viable virus within two days, and environmental dust made up a large fraction of the airborne particulates they emitted. The authors proposed that contaminated dust could serve as a vehicle for airborne transmission, and demonstrated transmission consistent with that mechanism [9].

For an air monitor, aerosolized fomites are both an opportunity and a complication. They mean that viable virus can be present in particle fractions that are not obviously "respiratory", and that a sampler collecting dust may recover virus that never travelled as a droplet. They also mean that a positive signal cannot be assigned to a particular route without additional information — the same limitation that the 2009 H1N1 literature ran into.

3.4 What remains uncertain

Three uncertainties should be carried forward. The first is the relative contribution of routes: for the 2009 H1N1 pandemic virus, transmission resembled seasonal influenza, but the shares attributable to aerosols, droplets and fomites could not be resolved [2]. The second is the infectious dose by the airborne route, which is difficult to establish experimentally and is often assumed rather than measured in modelling studies. The third is the behaviour of the virus in real indoor environments, where filtration, air exchange, temperature gradients and occupancy interact in ways that chamber experiments do not reproduce.

3.5 What this means for a detection device

The evidence points to a design that samples the sub-5 µm fraction deliberately, accounts for humidity-driven changes in particle size, and reports results in a way that does not overstate what a positive signal proves. Because one cough feeds both routes, a device that monitors only large droplets will systematically under-sample the virus that travels. Because aerosolized fomites exist, a device that collects all suspended particles should expect non-respiratory material in its samples and should be designed to tolerate it.

4. The virion: structure and genome

Everything a detector must recognise is contained in a particle roughly a hundred nanometres across, wrapped in a lipid membrane and studded with two glycoproteins. The structure matters for detection for a simple reason: what a recognition element can reach is determined by what the particle presents on its surface, and how densely and irregularly it presents it.

4.1 Morphology and pleiomorphy

Cryo-electron tomography of 110 virions, reconstructed from six tomograms, found that the majority were spherical, defined as an axial ratio below 1.2 (78 of 110 in the original classification), with the remainder oval, kidney-shaped or elongated [10]. The particles were sorted into five morphological classes, of which one comprised the elongated forms and the other four subdivided the spherical population according to features such as the presence of a discernible matrix layer.

Pleiomorphy is not a curiosity. It means that any measurement of "the size of an influenza virion" is a distribution rather than a number, and that a recognition surface must tolerate variation in curvature and in the spacing of glycoproteins. It also means that the virion is not a rigid object: the same genome can be packaged in particles that differ in shape and, presumably, in aerodynamic behaviour.

4.2 Genome organisation and conserved sequences

The genome consists of eight negative-sense single-stranded RNA segments encoding PB2, PB1, PA, HA, NP, NA, M and NS. Each segment is organised into three parts: untranslated regions at both ends, a packaging signal, and the coding region [11]. The untranslated regions contain the highly conserved sequences known as uni-12 and uni-13 (12 and 13 nucleotides respectively), which are shared across the eight segments and across influenza A strains — the reason they are used as universal primers in diagnostic RT-PCR.

The same literature records the distinction between antigenic drift, the gradual accumulation of mutations in the surface protein genes, and antigenic shift, the reassortment of whole segments between strains. Drift produces the seasonal pattern of immune escape; shift produces the antigenic novelty associated with pandemics. For a detector, the practical consequence is that any recognition element aimed at a variable epitope will lose sensitivity over time, and that assays targeting conserved regions trade specificity for stability.

4.3 Surface glycoproteins and their arrangement

Two glycoproteins dominate the surface: haemagglutinin (HA), which binds sialic acid receptors, and neuraminidase (NA), which cleaves them. A third, the M2 ion channel, is far less abundant but is the target of the adamantane drugs discussed in Chapter 9. In tomograms the glycoprotein spikes are packed closely but irregularly, with an average centre-to-centre spacing of about 11 nm; both HA and NA extend radially and terminate in bulbous heads, with NA sitting slightly higher than HA [10].

The density matters more than the individual protein. Antibodies and engineered receptors must find an epitope within a crowded field of glycans, and glycan shielding varies between strains and between

passage histories. A recognition element that works well on purified HA may perform differently on an intact virion, where the same epitope is presented at a different angle and with different neighbours.

4.4 What this means for a detection device

The virion presents a small, irregular, densely glycosylated surface. Recognition elements should be selected against intact particles rather than purified protein where possible, and the assay should be designed to tolerate a distribution of particle sizes and shapes. Where the target is a conserved sequence rather than a surface epitope — as in nucleic-acid detection — the accessible target is inside the particle, which makes lysis an explicit design step rather than an afterthought.

5. Environmental persistence

A virus that leaves the body is exposed to conditions it never met inside it: desiccation, ultraviolet light, temperature swings, reactive gases and surfaces of every kind. How long it remains infectious under those conditions determines whether an airborne detector is measuring a live threat or a molecular residue. This chapter brings together what the literature measures, and separates two things that are often conflated — how long the virus stays infectious, and how long its RNA can be detected.

5.1 Survival in air: ageing experiments

The most direct evidence comes from aerosol ageing experiments. Virus is nebulised into a closed exposure chamber, held for a defined interval, then re-sampled and assayed for infectious virus and for viral RNA. In one such study, all tested strains retained infectious virus after 15 and 30 minutes of ageing [12]. The study measured both plaque-forming units and RNA copy numbers, which allowed the authors to express loss of infectivity as a rising RNA-to-PFU ratio — a measure of how much of the detectable material was no longer infectious.

The humidity effect in that study was strain-specific rather than uniform. An H3N2 variant (Indiana/8) was more stable at 70% relative humidity; an H7N9 strain (Anhui/1) was least stable at 50% and more stable at 20% and 70%; and an H5N1 strain (Thai/16) showed no significant difference across humidities [12]. This is an uncomfortable result for anyone hoping for a single environmental rule. It implies that a humidity-based prediction developed for one subtype may not transfer to another, and that statements about "influenza stability" must specify the strain.

A second line of work complicates the picture further. When infectivity was measured across a range of relative humidities in aerosols and droplets, it was retained rather than sharply lost at any particular humidity [13]. Read together with the strain-specific results above, the literature supports a weaker claim than is often repeated: humidity modulates stability, but it does not act as a switch, and its effect depends on the strain and on the composition of the medium the virus is suspended in.

5.2 Humidity, temperature and the size of the airborne phase

Humidity acts on airborne virus through two routes that are easy to confuse. The first is biological: it changes the rate at which the virus loses infectivity. The second is physical: it changes the size of the droplet in which the virus travels, and therefore how long that droplet stays aloft. The physical route is often the larger of the two. Below about 64% relative humidity, a 10 µm droplet evaporates to roughly 1.9 µm, and its settling time rises from about eight minutes to more than three hours [8].

Temperature enters through both routes as well. It raises the rate of thermal denaturation of proteins and nucleic acid, and it determines how much water the air can hold, which in turn determines evaporation. Because absolute humidity and temperature are physically collinear — high absolute humidity can only occur at high temperature — studies that vary one while claiming to hold the other constant are difficult to interpret. The modelling literature therefore argues for treating relative humidity and temperature as a pair [8].

The practical implication for a measurement is that an indoor environment cannot be characterised by a single number. A room at 30% relative humidity and 18 °C is not equivalent to one at 30% and 26 °C, even though a hygrometer reports the same value. If a detection system is to be interpreted against environmental context, that context needs at least temperature and relative humidity, and ideally air exchange as well.

5.3 Surfaces and fomites: weeks of RNA, days of infectivity

The surface literature is unusually clear because one study followed both infectivity and RNA over seven weeks. Five H1N1 strains were deposited on cotton, microfibre and stainless steel and kept in dark and light conditions. Viable virus was recovered for up to two weeks from stainless steel and one week from cotton and microfibre, while PCR detection continued for the entire seven-week study with less than one log10 reduction [14]. The time to a 99% reduction in titre was 17.7 hours on cotton, 34.3 hours on microfibre and 174.9 hours on stainless steel. A separate study of the pandemic 2009 virus reported infectious virus for 48 hours on wood, 24 hours on stainless steel and plastic, and 8 hours on cloth [15]. The shorter windows are most plausibly a matter of inoculum and detection method rather than of a different intrinsic stability.

Two features of this result deserve emphasis. First, no strain differences were observed, which suggests that persistence on fomites is governed more by the material and the environment than by the isolate. Second, the divergence between the two readouts is not a small discrepancy — it is a difference of weeks. A surface that is PCR-positive may have been non-infectious for a month. Any monitoring programme that reports nucleic acid as though it were a proxy for infectious virus will misstate risk in a predictable direction.

5.4 What "stability" means when the readout changes

The persistence literature is often summarised as a set of half-lives, but a half-life is only meaningful with respect to a defined assay. Infectivity decays on a scale of hours to days; RNA decays on a scale of weeks; antigen detectability decays somewhere in between and depends on the assay. The three are not interchangeable, and the choice among them is a statement about what a study — or a device — claims to know.

5.5 What this means for a detection device

Three design consequences follow. The first is that environmental conditions must be recorded alongside the signal: without temperature, relative humidity and (where possible) air exchange, a reading cannot be placed on the stability curves reported here. The second is that the readout must be chosen deliberately. A device that detects RNA answers "was the virus here"; a device that detects infectious particles answers "is the virus still a threat". Both are useful, but they support different decisions. The third is that surface and air measurements answer different questions and should not be substituted for one another.

6. Entry, receptors and host range

The virus must do three things before it can replicate: find a cell whose surface it can bind, get through the mucus that covers that surface, and fuse its membrane with the cell membrane. Each step depends on the same two proteins — HA and NA — working in balance, and each step is a potential point of failure when the virus crosses into a new host. For a detector built on a receptor or an antibody, this chapter also defines what the recognition element is actually binding to.

6.1 Sialic acid linkage specificity

Influenza HA binds sialic acid, but not all sialic acids are equivalent. Human-adapted viruses bind α2-6-linked sialic acid, while avian viruses prefer α2-3 linkages. This distinction has been treated for decades as a central element of species specificity, and it remains the clearest molecular marker of host adaptation [16]. The distribution of the two linkages along the human respiratory tract — with α2-6 more prominent in the upper airway and α2-3 in the lower — is one reason why human viruses replicate efficiently in the nasopharynx while avian viruses more often cause severe lower-tract disease when they do infect people.

The specificity is not absolute. H7N9 viruses that have infected people show partial binding to α2-6 sialic acid, which may partly explain their frequent zoonotic transmission; H5N1 viruses remain α2-3-preferring, and substitutions at positions 226 and 228 are among the changes that switch specificity and are associated with airborne transmission [16]. These are not laboratory curiosities: they are the mutations that surveillance programmes look for when assessing pandemic risk.

6.2 The HA–NA–receptor balance

HA attaches the virus to sialic acid; NA removes sialic acid. The two activities oppose each other, and the balance between them governs how the virus moves. If HA binds too strongly, the virus is trapped in mucus or on the cell surface; if NA is too active, the virus cannot attach long enough to find a productive entry site. The literature on viral motility describes the result as a kind of rolling motion: low-affinity multivalent binding lets the particle explore the surface, while NA trims receptors locally to allow it to move.

The respiratory mucus layer is a particularly hostile environment for this process. It is rich in sialylated mucins that act as decoys, capturing virions before they reach the epithelium. Only continuous NA activity allows the virus to follow a receptor density gradient through the mucus toward the cell surface [17]. At the end of the cycle, NA has the opposite role: it cleaves receptors on the surface of the infected cell so that newly assembled virions are not recaptured by their own HA before they can spread.

Cross-species transmission requires rebalancing this system for a new host. The sialoglycan repertoire differs between species in type, density and distribution, so a balance tuned in one host may fail in another. Adaptation can occur on either side: HA receptor-binding mutations such as Q226L and G228S, or NA changes such as loss or modification of the second sialic acid binding site, altered catalytic activity, or changes in stalk length. In H7N9, NA changes at the second binding site appear to have preceded the HA Q226L substitution, which suggests that NA adaptation can come first [17].

6.3 Fusion and the activation pH

Entry ends with membrane fusion. HA is synthesised as an uncleaved precursor, HA0, which cannot fuse until a host protease cuts it into HA1 and HA2. Cleavage is therefore a host-dependent step and a determinant of tissue tropism: viruses whose HA is cleaved by proteases restricted to the respiratory tract cause local infection, while those with cleavage sites recognised by ubiquitous proteases can spread systemically [18].

After cleavage, acidic pH in the endosome triggers a large conformational change. The stem loop unfolds into a long helix, and the fusion peptide is thrust toward the target membrane — the movement the literature describes as a "harpoon" [18,19]. HA activation pH values typically fall between 4.8 and 6.2, and the value differs between subtypes and hosts, and the range itself is a virulence trait [20]. Caffrey and Lavie note that the extracellular pH of the nasal cavity is mildly acidic — around 6.3 in healthy adults and 5.9 in children — which places some activation thresholds close to the pH the virus meets before it ever enters a cell.

6.4 What this means for a detection device

For a receptor-based recognition layer, the lesson is that binding affinity alone is insufficient. The natural system uses low-affinity multivalent engagement plus enzymatic release to achieve specificity and mobility; a synthetic receptor that binds too tightly may be no better than one that binds too weakly, because both fail to discriminate between the target and the decoys. For an antibody-based layer, the lesson is that the epitope must be accessible in the intact particle, at the glycan density the virion actually presents.

7. Detection: what we measure and what it means

Every detection method answers a narrower question than it appears to. A plaque assay answers "how many particles can still initiate infection in this cell line". An RT-qPCR answers "how many copies of this sequence are present". An antigen test answers "is this epitope available to this antibody in this buffer". None of them answers "is this person infectious" directly, and the gap between the questions is where most misinterpretation occurs.

7.1 Infectivity assays

Infectivity is measured by exposing susceptible cells to a dilution series and counting the resulting plaques or infected wells. The result is expressed as plaque-forming units or a tissue-culture infectious dose. These assays are the closest available proxy for the biological question, but they are slow — days for influenza — and they depend on the cell line, the passage history of the virus and the conditions of the assay. Two laboratories can obtain different titres from the same sample without either being wrong.

7.2 Nucleic acid tests

RT-qPCR detects a conserved sequence, usually in the matrix gene or the conserved untranslated regions described in Chapter 4. It is fast, sensitive and quantitative over a wide range, which is why it became the reference method during the COVID-19 pandemic. Its limitation is conceptual rather than technical: it detects the sequence, not the particle. On surfaces, RNA remained detectable for seven weeks while viable virus was recovered for one to two [14]. For airborne monitoring, the same asymmetry means that a PCR-positive air sample may reflect virus that has been inactivated by desiccation or ultraviolet light.

7.3 Antigen detection and the rapid-test reference point

Lateral-flow antigen tests occupy the point-of-care niche between infectivity assays and nucleic acid tests. They are fast and cheap, and they detect protein rather than sequence, which in principle brings them closer to a biological question. In practice their sensitivity is lower than that of nucleic acid tests, and their performance depends heavily on when in the illness the sample is taken, because antigen load peaks early. The clinical literature on influenza repeatedly makes the same point about the whole category: a negative rapid test does not exclude infection, and a positive one is a screening result that requires confirmation.

7.4 The gap between signal and state

The distinction between nucleic acid and infectivity is the single most important thing this document has to say to a detection team. It is not a technical caveat; it determines what a device can claim. A device that reports "influenza A RNA detected" is making a claim about presence. A device that reports "infectious influenza A present" is making a claim about a biological state, and that claim requires either an infectivity assay or a validated surrogate for one. Between the two lies a design space in which most biosensors, including ours, actually operate.

7.5 Air sampling and the size question

Air sampling inherits the physics of Chapter 3. Particles of different sizes are captured with different efficiencies by different samplers, and the size distribution of infectious particles changes as droplets evaporate. A sampler that is validated against laboratory aerosols of a single size may behave differently in a room where the distribution is broad and the humidity is low. Because the same cough produces particles from a few micrometres to tens of micrometres [8], the sampler has to make an explicit choice about which part of that distribution it represents. Direct measurements of airborne virus in particles from human coughs confirm that infectious material is present in the aerosol fraction, although in small quantities [21], which is why a sampler has to concentrate a large volume of air rather than simply wait.

7.6 What this means for a detection device

A detection team should state, in advance, which of the three questions its device answers — presence, infectivity, or exposure — and should design the readout, the sampling and the reporting around that choice. This document records what each method can and cannot support, so that the choice can be defended rather than assumed. This is also the point at which our own project had to become explicit: our biosensor reports a cellular response to viral antigen, which is closer to presence than to infectivity, and our claims have to be worded accordingly.

8. Antibodies and recognition elements

Antibodies are the best-characterised recognition elements for influenza, and the literature on them is also the most instructive for anyone engineering an alternative. It shows which parts of the virion the immune system finds accessible, why some antibodies are broad and others narrow, and how the virion's own architecture limits what any binder can reach.

8.1 Head versus stem

The globular head of HA is strongly immunogenic and elicits potent neutralising antibodies, but it is also the most variable part of the protein, so antibodies against it tend to be strain-specific. The stem is more conserved and antibodies against it are broader, though they are usually less potent in neutralisation assays. The practical consequence is a trade-off that recurs across detection as well as therapy: binders against variable regions give strong signal against the strain they were raised against and fail against drift variants; binders against conserved regions give weaker but more durable recognition [22].

8.2 Broadly neutralising antibodies and hidden epitopes

A subset of antibodies neutralises viruses across subtypes, and their structures have been mapped in some detail — for example CR6261 and F10 against group 1 HA, CR8020 against group 2, and CR9114, FI6v3 and MEDI8852 across groups [22]. A further class binds the HA head but not at the receptor-binding site; the epitope these antibodies recognise is hidden in the pre-fusion form of HA and becomes exposed only after the protein rearranges. That observation is directly relevant to detection, because it means the accessibility of an epitope depends on the conformational state of the particle, which in turn depends on its environment.

8.3 How epitopes are mapped

Residue-level epitope maps come mainly from X-ray crystallography of antibody–antigen complexes. The method requires purified antibody and antigen, formation and often purification of the complex, and successive rounds of crystallisation — which is why it produces very precise contact maps for a small number of antibodies rather than broad coverage. Nuclear magnetic resonance can give finer information about contacts in solution, but it requires a prior structure of the free antigen and is not commonly applied at this scale [23]. Biosensor-based methods offer a middle path: they map epitopes by measuring how panels of antibodies bind to surface-immobilised HA and NA, without crystallising each complex, at the cost of lower structural resolution [24].

8.4 What this means for a detection device

Three practical points follow. First, the choice between head- and stem-directed binders is a choice about strain coverage, and should be made with the intended use in mind. Second, epitope accessibility in an intact virion is not the same as accessibility in purified protein, so binding should be characterised on particles. Third, because some epitopes are conformation-dependent, sample handling that alters pH or osmolarity can change the signal without changing the amount of virus present.

9. Antivirals, resistance and vaccines

The therapeutic landscape matters to a detection project in two ways. It defines what a positive result would change — whether there is an action to take — and it illustrates how quickly a virus can escape a single-target intervention, a lesson that applies to recognition elements as much as to drugs.

9.1 Drug classes and their targets

Three classes are approved and in use. The M2 ion-channel inhibitors, amantadine and rimantadine, block the proton channel required for uncoating. The neuraminidase inhibitors — oseltamivir, zanamivir and peramivir — prevent release of progeny virions. Baloxavir inhibits the cap-dependent endonuclease of the polymerase complex, blocking transcription [1]. The three classes act at different points in the life cycle, which is why the literature on resistance treats them separately.

9.2 Resistance and the next generation

Resistance mutations are well mapped. For oseltamivir they include NA H275Y, R292K and N294S; oseltamivir-resistant strains generally remain sensitive to zanamivir. For baloxavir, substitutions at PA I38 and E23 have been identified in treated patients. M2 resistance is widespread enough that the adamantanes are no longer recommended for influenza A in many settings, and it has driven the search for newer channel inhibitors [1].

The next generation includes agents at different stages of development: laninamivir and its prodrug, favipiravir, pimodivir, and newer polymerase and channel inhibitors, alongside combination strategies and host-directed approaches [1]. The recurring theme is that single-target drugs select resistance quickly, and that combinations and higher genetic barriers are the standard responses.

9.3 Timing of treatment

Clinical benefit depends on when treatment starts. In the 24-hospital US network described in Chapter 2, adults hospitalised with influenza who began oseltamivir on the day of admission were less likely to progress after admission (11% versus 21% for a one-level escalation) and had lower odds of ICU admission, with an adjusted odds ratio of 0.25 [6]. The study also reported a shorter length of stay and lower odds of acute vasopressor use or new kidney replacement therapy. These are modest effects by the standards of trial medicine, but they are the effects that exist, and they decay with delay.

9.4 Vaccines and their limits

Vaccination remains the principal population-level intervention, and its limits are well documented. Reported US effectiveness against medically attended influenza ranged from about 10% to 60% between 2004 and 2021, varying with strain match, population and season [1]. The variation is not a failure of the vaccines; it is a consequence of a virus whose surface antigens change faster than vaccine production cycles, and of an immune response that is shaped by prior exposure to related strains.

9.5 What this means for a detection device

Two implications stand out. First, detection has clinical value only if it shortens the interval to treatment, which makes the speed and accessibility of the assay as important as its sensitivity. Second, the escape dynamics that produce drug resistance also produce antigenic drift, so any recognition element will need periodic reassessment against circulating strains. A device designed around a single binder is a device with a shelf life measured in seasons.

10. The clinical picture and its comparisons

A biosensor is not a clinical instrument, but it lives in a clinical world. How influenza presents, who becomes severely ill, and how the disease differs from the infections it is usually confused with determine what a detection result would be used for. This chapter summarises those comparisons.

10.1 Lessons from the 2009 H1N1 pandemic

The 2009 pandemic produced a large, well-characterised clinical literature in a short time. Transmission appeared similar to seasonal influenza, though the relative contributions of small-particle aerosols, large droplets and fomites could not be resolved [2]. Severe disease was not confined to the usual risk groups: approximately one quarter to one half of hospitalised or fatal cases had no reported coexisting medical condition. Underlying conditions that increase risk in seasonal influenza — chronic lung disease, immunosuppression, pregnancy, obesity — also increased risk here, but they did not account for the majority of severe outcomes.

For a detection project, the implication is about where to deploy. If severe cases arise unpredictably across the population, then monitoring only high-risk settings — hospitals, care homes — will miss the environments where transmission is actually accumulating. Schools, workplaces and transport are not clinical settings, but they are where the next severe case is most likely to have been exposed.

10.2 Neurological complications

Influenza is an under-recognised cause of central nervous system dysfunction. Febrile seizures are common in infants and children with influenza and are an important cause of hospitalisation; influenza-associated acute encephalitis and encephalopathy are rarer but considerably more serious [3,4,5]. The aetiology of febrile seizures is multifactorial — fever itself is a trigger, and the height of fever correlates with risk more than the rate of its rise — which is one reason why the neurological picture varies so much between children with similar infections.

This material is included for a reason that is easy to miss: it shows that the clinical significance of influenza is not captured by respiratory symptoms alone. A device that detects virus in a paediatric ward is not diagnosing encephalitis, but it can identify the exposure that precedes it.

10.3 Influenza and COVID-19

SARS-CoV-2 and influenza are different pathogens with overlapping presentation. Both are transmitted primarily by respiratory droplets, which is why the non-pharmacological measures adopted during the COVID-19 pandemic also reduced influenza transmission. Beyond that, the differences matter more than the similarities. Influenza can be treated with neuraminidase inhibitors or a cap-dependent endonuclease inhibitor, neither of which has activity against SARS-CoV-2; remdesivir, used for COVID-19, has no established role in influenza [25].

The clinical courses also differ. Patients with influenza typically experience their most severe symptoms in the first week of illness, whereas COVID-19 may run a longer course with a peak in the second or third week [25]. The consequence for testing is that timing assumptions cannot be transferred from one

disease to the other: a test strategy optimised for early peak viral load in influenza may miss COVID-19, and vice versa. The same review notes that correctly identifying the virus has infection-control consequences — isolation, return to work or school, contact tracing — that differ between the two.

10.4 Influenza and respiratory syncytial virus

Clinical heuristics treat fever as a hallmark of influenza and wheezing as typical of respiratory syncytial virus (RSV). We examined how well these associations hold in the published literature by pooling the available studies [26]. The useful conclusion for a detection project is not that the heuristics are wrong, but that they are probabilistic: symptoms narrow the differential but do not identify the pathogen, which is precisely the gap that a specific test fills.

10.5 What this means for a detection device

The clinical picture sets the requirements: the device must discriminate influenza A from the infections it resembles, it must produce a result fast enough to change management, and it must be interpretable by someone who is not a virologist. None of these requirements is met by sensitivity alone.

11. Prevention and control: where environmental monitoring fits

Control of influenza rests on vaccination, antivirals, surveillance and non-pharmaceutical measures. Environmental monitoring is not usually listed among them, and this chapter argues that it belongs at the margin — not as a replacement for clinical surveillance, but as a way of seeing transmission before cases present.

11.1 Surveillance and its blind spots

Influenza surveillance measures incidence in several ways: medically attended illness, laboratory-confirmed cases, serological estimates of cumulative infection, and increasingly, syndromic data such as calls to telehealth services [27]. Each measure has a characteristic blind spot. Laboratory-confirmed counts depend on testing practice, which changes over time and between settings; serology captures infections that were never tested but lags behind them; syndromic data are sensitive but non-specific.

None of these methods observes the environment in which transmission occurs. They observe people who have already been infected, usually after they have become ill enough to seek care. The interval between infection and reporting is the interval in which transmission happens.

11.2 Non-pharmaceutical measures and the physics behind them

Ventilation, filtration and humidity control are recommended for indoor spaces on general grounds, and the physics in Chapter 3 explains why. Increasing air exchange removes airborne particles before they are inhaled; filtration removes them from recirculated air; and humidity changes both the size of droplet nuclei and the rate at which virus loses infectivity. The evidence is strongest for the first two, because the effect of ventilation on particle concentration is direct and measurable, and weaker for humidity, because its biological effect is strain-dependent [12] and its physical effect interacts with temperature [8].

This asymmetry is worth stating plainly, because humidity recommendations are often presented with more confidence than the evidence supports. What the literature supports is that low humidity favours evaporation and longer airborne residence, and that its effect on infectivity varies by strain. It does not support a single target humidity that works for all settings and all strains.

11.3 Why environmental monitoring complements clinical surveillance

Environmental monitoring measures something clinical surveillance cannot: the amount of virus in a space, independent of whether anyone has yet become ill enough to be tested. It has three potential uses. It can provide an early warning in settings where clinical testing is sparse. It can support decisions about ventilation, filtration and occupancy with measurements rather than assumptions. And it can be used to evaluate interventions, because a change in air concentration is observable before a change in case counts.

The same chapter of the literature that makes this argument also limits it. A positive air sample does not identify who is infected, and a negative one does not prove a room is safe, because sampling is intermittent and the infectious dose by the airborne route is not established. Environmental monitoring is

a complement to clinical surveillance, not a substitute for it — a conclusion our own project has had to state explicitly.

12. How this document was built

The value of a compilation depends on its method being visible. This chapter describes how the questions were selected and answered, what was included and excluded, and where the approach falls short of a systematic review.

12.1 A question-driven structure

The document is organised around twenty-one questions, each of which came from a real conversation or a real design decision. They are grouped into five families: survival and persistence; size, morphology and transmission; entry and host range; detection and measurement; and clinical and intervention questions. A question-driven structure was chosen over a textbook structure because it matches how the material is used — a team arrives with a question, not with a syllabus.

12.2 Search and appraisal

Each question was searched in the primary literature, with reviews used to locate studies and primary papers used to verify numbers. Priority was given to studies that reported quantitative data, that specified the strains and conditions used, and that have been cited consistently since publication. Where a claim appeared in several reviews but traced back to a single study, we cite the study rather than the reviews. Where reviews disagreed, both positions are recorded.

The appraisal was not formal. We did not apply a risk-of-bias tool or a predefined quality score, and we did not register a protocol. What we did do was check every number against the paper that reported it, record the conditions under which it was obtained, and mark the cases where a review and its cited source did not agree. In one such case a figure widely repeated in secondary sources could not be confirmed in the primary paper, and we say so.

12.3 What each chapter contains

Every chapter follows the same pattern: the question; the short answer; the evidence, with numbers and the conditions under which they were obtained; what that evidence means for a detection device; and the references. The uniform structure is deliberate — it allows a reader to check a claim in one place, and it allows new material to be added without redesigning the document.

12.4 Limitations

Four limitations should be stated. First, the document is selective: it covers the questions we needed to answer, not the whole of influenza biology. Second, it is time-bound: the literature cited ends in 2026, and some of the numbers — vaccine effectiveness in particular — change every season. Third, it is a secondary compilation: we read and assessed the primary literature but did not repeat any experiment. Fourth, the appraisal was informal, so what follows should be read as curated evidence rather than as a systematic review.

13. How this document can be reused

A contribution is only a contribution if someone else can use it. This chapter describes the three things we made — the knowledge base, the discussion format and the workshop record — and how each can be taken up by another team.

13.1 Reusing the text

The sections are written to be lifted whole. A team writing a wiki page on airborne detection can quote the persistence section together with its sources; a team completing a safety form can use the transmission and inactivation material to justify containment choices; a team preparing a presentation can use the transmission and persistence material, with its citations. In each case the requirement is the same: keep the citation with the claim.

13.2 The discussion format: APiC question cards

Our Human Practices work produced a second, less obvious contribution: a format for peer discussion. At the APiC project exchange we replaced the usual sequence of project presentations with structured question cards built around iGEM award tracks. Each card directed discussion through five dimensions — value positioning, design rationale, analysis of previous projects, critical self-evaluation, and future optimisation — so that teams spent their time examining why a design was appropriate rather than describing what it aimed to do.

The format proved reusable because it does not depend on shared subject matter. Teams working on different biological questions can use the same cards, because the questions are about reasoning rather than about pathogens. The cards and the process are documented so that future organisers can adapt them to other award tracks, exchange events and review sessions.

13.3 The workshop record: making experimental experience shareable

The third contribution came out of an online yeast workshop that brought together CAU-China, NEFU-China, SZPU-China, Keystone, NKU and SHSID. The discussion covered transformation efficiency, gene knockout, selection stability, reporter leakage, membrane-protein expression and localisation, product formats and biosafety. Much of what made it useful was not the methods themselves but the troubleshooting attached to them: which failure modes were recognised, which controls distinguished them, and what remained unresolved.

Two lessons from that discussion are recorded in the workshop write-up. The first is a methodological caution: a mutation at a target site does not by itself establish that Cas9 cleavage caused it, so controls and independent genotype verification are needed before a knockout is claimed. The second is that methods shared by other teams should be preserved with their attribution — CAU-China's combination of fluorescently tagged receptors, FM4-64 membrane staining and confocal microscopy, and the alginate microsphere and freeze-dried colour-card formats discussed by Keystone and NKU, are recorded as theirs. None of the methods discussed originated with us; the value of the record lies in the exchange itself and in keeping each method attached to the team that described it.

13.4 What we learned about sharing knowledge

Three things became clear across these activities. Structured questions produce better exchanges than presentations, because they force specificity. Documented troubleshooting is more valuable than documented success, because most teams fail in similar ways. And a contribution is strongest when it preserves attribution, because that is what allows the next team to build on it rather than repeat it.

14. Conclusions

What the literature supports, in one paragraph: influenza A remains infectious in air for at least tens of minutes and, under dry conditions, in droplet nuclei that stay aloft for hours; humidity modulates both the physical lifetime of those nuclei and the biological lifetime of the virus, with effects that depend on strain; viable virus persists on surfaces for days to weeks while its RNA remains detectable for far longer; transmission occurs by droplets, aerosols and contaminated dust, and the boundaries between these routes are conventions rather than mechanisms; and detection methods differ in what they measure, with the gap between nucleic acid and infectivity being the most consequential for anyone building an environmental sensor.

What remains open: the infectious dose by the airborne route; the relative contribution of routes in real indoor environments; the behaviour of pleomorphic virions in samplers designed around spherical particles; and the question that motivated this work in the first place — how to interpret a positive environmental signal when the virus it detects may no longer be infectious.

What we would add next: material on air-sampling instrumentation and on the validation of environmental signals against clinical outcomes, both of which were outside the scope of what we could verify within the project. We would also add a version of the discussion format that other teams can run without our facilitation, so that the contribution outlives the team that made it.

This document is offered in that spirit. It is not a claim that we have answered these questions, but a record of how far the evidence goes, where it stops, and what a team can reasonably do with it in the meantime.

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